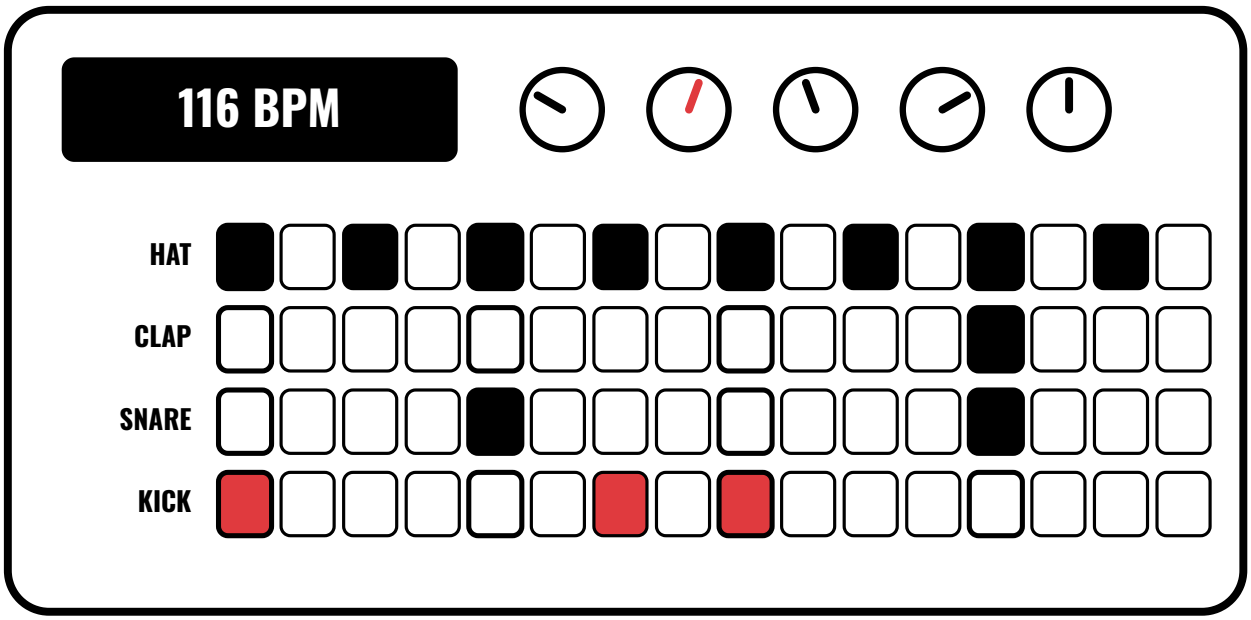
HUDSON BEAT



DRUM PROGRAMMING IN LOGIC PRO

BEAT FROM NOTHING

Digital Music Production 1 · Unit 1 · teacher notes, student pages, /25 rubric, intro deck

GRADES 9–12 · 4 DAYS · 21 PAGES + 15-SLIDE DECK

BEAT FROM NOTHING

How to use this resource

START HERE

Teacher

WHAT THIS IS ·

A complete 4-day drum-programming unit for high school music technology. Students build an original drum beat from nothing in Logic Pro's Drum Machine Designer: no loops, no pre-made grooves. They name the kit, hear how loops work, copy one famous pattern, then build and mix their own 16-measure beat in AABA form and present it.

This is the first unit of my Digital Music Production 1 course, the one I teach every September. It works just as well on its own.

QUICK START ·

1. Read the overview and pacing (pages 3–5) and the teacher notes (pages 8–9).
2. Check the setup list (page 4) on one student computer.
3. Program the three listening examples into Logic from the key (page 19): about 10 minutes.
4. Open the intro deck and cue up Billie Jean (Michael Jackson) on YouTube or Spotify. No audio is included.
5. Print or post the student pages (pages 10–18).

THE INTRO DECK ·

You present it: slides 1–8 on Day 1 (hook, what a loop is, beats and measures, the kit, the grid) and slides 9–15 to launch the project on Day 2 (worked example, Logic steps, the project, deliverables, rubric). Every slide has speaker notes.

Fonts: the deck uses Oswald and Source Sans 3 (free on Google Fonts, built into Google Slides). If PowerPoint shows a different font, install them or use the PDF copy, which shows the intended look.

WHAT'S INSIDE ·

PART	PAGES	USE
Teacher pages	pages 2–9	How to use, overview, setup, pacing, day plans, teacher notes
Student pages	pages 10–18	Day 1 post, worksheets, patterns, project sheet, vocabulary, checklist, rubric
Answer key + example notes	pages 19–20	Kit labels, listening answers, a worked example beat and mix
Intro deck (separate file)	deck	15 slides, .pptx + PDF, speaker notes on every slide

DIGITAL OR PRINT ·

Every student page prints in black and white. To go digital, post the PDF pages to your class page; students can fill in the listening grid and checklist with any PDF annotation tool.

MAKE IT FIT YOUR CLASS ·

- **Short periods or a slower group?** Split Day 3 into two build days (A section; then B, AABA and mix). That is my original 5-day pacing.
- **Need it simpler or harder?** See the variations on page 9.
- **Feedback board:** I use Padlet; any discussion board, LMS forum or paper gallery walk works.

NO COPYRIGHTED MEDIA

Songs are named by title and artist only. Play them from your own YouTube or Spotify access. The drum grids are simple counting charts for teaching; no audio, video, lyrics or album art are included.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Unit overview

OVERVIEW

DMP1 · Unit 1

WHAT STUDENTS MAKE ·

An original drum beat built entirely from scratch in Drum Machine Designer: no loops, no pre-made grooves. Students copy one famous pattern, then build an original one inspired by it: 16 measures in AABA form, mixed with volume and panning.

WHAT STUDENTS LEARN ·

Kick, snare, and hi-hat by name, sound, and function. Subdivision: eighth notes, sixteenth notes, and why they feel different. Why repetition is a musical tool, not laziness. The satisfaction of making something from nothing.

WHY IT COMES FIRST ·

This is the hook. But instead of 'drag a loop and feel like a producer,' the hook is authorship: 'I made that sound myself and I understand why it works.' Students who build a beat understand what is inside every loop they will meet for the rest of the year. Constraints force musical decisions.

MIXING CONCEPT: VOLUME & PANNING ·

The first mixing lesson is simple: not everything should be the same volume. The kick should feel grounded, the hi-hat should sit above it, the snare should cut through. Turning something down to make something else feel louder is the most transferable mixing lesson there is.

GOALS ·

- Identify the kick, snare, and hi-hat by name, sound, and function
- Build an original drum beat using Logic Pro's Drum Machine Designer
- Understand how volume and panning affect the feel of a mix
- Experience the creative process of building something musical from scratch

AT A GLANCE ·

Grades	9–12, beginners (no music reading needed)
Length	4 class periods of about 53 minutes: 2 learning, 1 work, 1 presenting
Type	Structured project
Software	Logic Pro: Drum Machine Designer, pattern regions, Step Sequencer (16 and 64 steps), volume and pan
Product	16-measure AABA drum beat, MP3 + compressed project, presented with three feedback questions
Assessment	5-category rubric, /25 (page 18)

STANDARDS ·

Supports the National Core Arts Standards for Music (Music Technology, high school): generating and organizing musical ideas (MU:Cr1, MU:Cr2), refining work through feedback (MU:Cr3), presenting (MU:Pr6) and analyzing what is heard (MU:Re7).

A NOTE FROM ME

Some students will be frustrated that they can't just drag a loop. Hold the line. By the end of this unit they will feel more pride in four bars they built than in four minutes they assembled. That pride is the hook, and it lasts longer.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Setup and requirements

SETUP

Teacher

YOU NEED ·

- **A Mac with Logic Pro** for each student (Logic Pro X 10.5 or later, for the Step Sequencer and pattern regions).
- **Headphones** for every student.
- **Projector and speakers** for the intro deck and the listening examples.
- **Billie Jean (Michael Jackson)** from your own YouTube or Spotify access. Optional for comparison: the other two songs on the Famous Drum Patterns page.
- **A place to submit** an MP3 and a .zip (your LMS or class page).
- **A feedback board** for Day 4 (Padlet, an LMS discussion, or paper).

OPTIONAL ·

- **A small USB MIDI keyboard or pad controller** (an MPK Mini or similar) to audition sounds and play hits in. The project is built in the Step Sequencer, so a keyboard is never required.
- A short "parts of a drum set" video for Day 1 (search YouTube; any clear 3–5 minute video works).
- The free Learning Music beat-making lessons at learningmusic.ableton.com for students who want more.

BEFORE DAY 1 ·

- [ ] Open Logic on one student computer and make a pattern region (checks the version and the kits).
- [ ] Program the three listening examples from page 19 as three pattern regions (or three projects).
- [ ] Make a quick example beat of your own to play on Day 2.
- [ ] Test the projector sound and cue up Billie Jean.
- [ ] Print or post the student pages; set up the submission spot and the feedback board.

LOGIC QUICK REFERENCE ·

New drum track	Create a Software Instrument track and choose any kit under Electronic Drum Kit in the Library (Y). This loads Drum Machine Designer.
Pattern region	Control-click (right-click) in the track at bar 1 and choose Create Pattern Region. Double-click it to open the Step Sequencer.
Place hits	Click squares to turn hits on. Each row is one sound; add rows for more sounds. Pattern length goes up to 64 steps.
Cycle	Press C, then set the cycle area over bar 1 so it repeats while you work.
Tempo	Double-click the tempo in the control bar and type the BPM.
Tracks	Duplicate a track with Track › New Track With Duplicate Setting (Command-D). Change a sound from the Library.
Repeat	Option-drag a region to copy it, or select it and press Command-R to repeat it.
Volume and pan	Open the Mixer (X). Click the disclosure triangle on the Drum Machine Designer track to show each drum's own fader and pan knob.
Export	File › Bounce › Project or Section (Command-B), MP3. In Finder, Control-click the .logicx project and choose Compress.

Menu names are from current Logic Pro versions; older versions may differ slightly. Logic Pro is a trademark of Apple Inc.; it is named here only to describe what the unit needs. This resource is not affiliated with Apple.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Pacing guide · 4 days (2 learning / 1 work / 1 present)

PACING

Teacher

Four class periods of about 53 minutes: two learning days, one work day, one presenting day. Each day's full plan, minute by minute, is on pages 6–7.

DAY	FOCUS AND GOAL	MINUTE BLOCKS	SHOW
DAY 1 Learning	**Name the kit, hear the loop** Students can name every part of the kit, define a loop, and count beats and measures by ear. Nothing is built in Logic yet.	0–5 min · Hook 5–17 min · Parts of the drum set 17–35 min · Intro deck: loops 35–50 min · Drumbeat listening 50–53 min · Exit check	Intro deck slides 1–8
DAY 2 Learning + first time in Logic	**Copy a famous pattern** Every student can create a pattern region, cycle a measure, add and duplicate tracks, change an instrument and set the tempo, and has one famous pattern checked off. The project launches at the end of class.	0–5 min · Vocabulary review 5–15 min · First pattern region 15–23 min · Logic techniques 23–38 min · Copy a famous pattern 38–53 min · Launch the project	Intro deck slides 9–15
DAY 3 Work day: build and mix	**A, B, AABA, mix, submit** Project submitted by the end of class: 16 measures in AABA, every track's volume set, two tracks panned.	0–18 min · A section 18–30 min · B section 30–37 min · Arrange AABA 37–48 min · Mixing focus 48–53 min · Export and submit	Project sheet + checklist
DAY 4 Presenting	**Share and give feedback** Every student plays their beat, hears feedback, and gives specific feedback to two classmates.	0–5 min · Feedback questions 5–50 min · Presentations 50–53 min · Close	Rubric

NEED A FIFTH DAY?

Split Day 3: build the A section on one day; the B section, AABA and mix on the next. That is the 5-day version I have taught. Presentations then move to Day 5.

AABA AT A GLANCE



4 measures each · 16 measures · B = A with one change

STUDENT PAGES BY DAY ·

Day 1	Day 1 post (p. 10), Parts of the Drum Set (p. 11), Drumbeat Listening (p. 12), Vocabulary (p. 16)
Day 2	Famous Drum Patterns (p. 13), Project Sheet (pp. 14–15), Rubric (p. 18)
Day 3	Project Sheet steps 4–7, Submission Checklist (p. 17)
Day 4	Rubric for self-score and teacher score (p. 18)

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Day-by-day plans · Days 1 and 2

DAYS 1–2

Learning

DAY 1 · NAME THE KIT, HEAR THE LOOP

Learning · Intro deck slides 1–8

Goal: Students can name every part of the kit, define a loop, and count beats and measures by ear. Nothing is built in Logic yet.

0–5 **HOOK** · Play about 60 seconds of Billie Jean (Michael Jackson) from YouTube or Spotify and ask what students hear. What's repeating? What enters? Don't give them answers yet.

5–17 **PARTS OF THE DRUM SET** · Show a short parts-of-the-drum-set video, then students label the Parts of the Drum Set worksheet from the word bank. Review: every student names kick (bass drum), snare, hi-hat, toms, crash and ride without looking.

17–35 **INTRO DECK: LOOPS** · Define loop: a repeated musical idea that builds on what came before. Replay Billie Jean and name each looping instrument. Clap the pulse: each clap is a beat, 4 beats = 1 measure. Count how many measures the drums play alone, then map the intro layer by layer and show it as regions in Logic.

35–50 **DRUMBEAT LISTENING** · Play the three listening examples (programmed from the key on page 19). Students mark where each sound lands on the grid. Fill it in together, then compare: why does one feel different from another?

50–53 **EXIT CHECK** · Words you should own: kick, snare, hi-hat, toms, crash, ride, loop, beat, measure, bass.

DAY 2 · COPY A FAMOUS PATTERN

Learning + first time in Logic · Intro deck slides 9–15

Goal: Every student can create a pattern region, cycle a measure, add and duplicate tracks, change an instrument and set the tempo, and has one famous pattern checked off. The project launches at the end of class.

0–5 **VOCABULARY REVIEW** · Cold-call the parts of the kit. What is a loop? How many beats are in a measure? What made the three listening examples feel different?

5–15 **FIRST PATTERN REGION** · New project, Drum Machine Designer, then Control-click and Create Pattern Region. Students mess around: no requirements yet. Hands on the grid before anything is graded.

15–23 **LOGIC TECHNIQUES** · Demonstrate to the whole class: Cycle, add a track, duplicate a track, change the instrument on a track.

23–38 **COPY A FAMOUS PATTERN** · Students delete what they made, choose ONE pattern from the Famous Drum Patterns page and recreate it exactly. Every student shows you theirs; use that check-in to show them how to change the project tempo.

38–53 **LAUNCH THE PROJECT** · Hand out the project sheet and rubric and read the requirements together. Explain the grid: 16 squares = 16 sixteenth notes; every other square = eighth notes. Play the starting point (kick 1 and 3, snare 2 and 4, hi-hat in eighths), change one thing, and let them hear how far the feel moves.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Day-by-day plans · Days 3 and 4

DAYS 3–4

Work + present

DAY 3 · A, B, AABA, MIX, SUBMIT

Work day: build and mix · Project sheet + checklist

Goal: Project submitted by the end of class: 16 measures in AABA, every track's volume set, two tracks panned.

0–18 **A SECTION** · Students build a 4-measure A section that is theirs, not the famous pattern. Circulate. The prompt when a student is stuck: 'Can you nod your head to it? If not, what one thing would you change?'

18–30 **B SECTION** · Demonstrate: duplicate the A pattern region and change a single element. Redirect students who start from scratch. 'Play me your A and then your B. Can I hear the difference?'

30–37 **ARRANGE AABA** · A, A, B, A, four measures each, sixteen measures total. Check region lengths before anyone moves on.

37–48 **MIXING FOCUS** · Play a mix with everything at max, then pull elements down one at a time. Students set every track's volume and pan at least two. Remind: turn things down, not up.

48–53 **EXPORT AND SUBMIT** · Bounce an MP3, name it correctly, compress the .logicx, and submit both.

DAY 4 · SHARE AND GIVE FEEDBACK

Presenting · Rubric

Goal: Every student plays their beat, hears feedback, and gives specific feedback to two classmates.

0–5 **FEEDBACK QUESTIONS** · Students post three feedback questions to the class board (Padlet or similar) before class begins.

5–50 **PRESENTATIONS** · Each student plays their beat once or twice and shares their three questions. Classmates respond to at least 2 projects. Allow about 3–4 minutes per student.

50–53 **CLOSE** · 'Every single one of those beats was made from scratch. Nothing you heard was a pre-made loop.' Let that land.

CIRCULATING PROMPTS

- 'Can you nod your head to it? If not, what one thing would you change?'
- 'Play me your A and then your B. Can I hear the difference?'
- 'Can you clearly hear each drum sound? What could you turn down?'

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

WHAT THIS UNIT REINFORCES ·

This is the entry point for the course. It sets the tone by establishing authorship over assembly as the hook: students who build a beat from scratch understand what is inside every loop they will encounter for the rest of the year. The first two days are deliberately slow: students name the kit, learn what a loop actually is, and count beats and measures by ear before they are asked to build anything. The AABA form requirement plants the first formal seed: they meet the term on the project sheet and hear it in their own work before they can define it. The mixing concept (volume and panning) is introduced here in the simplest possible context: three sounds competing for space is the most accessible mixing problem you can give a beginner.

The Famous Drum Patterns activity on Day 2 is critical. Copying Billie Jean is not busywork: it trains the ear to recognize what 'correct' feels like at 116 BPM with a specific kick-snare-hi-hat relationship. Students who complete it properly have a reference point for everything they build afterward.

COMMON STUDENT STRUGGLES ·

The blank page problem: Some students will stare at the step sequencer and not know where to start. The Day 2 sequence solves this: let them mess around in a pattern region with no requirements first, then copy a famous pattern, then delete it and build something new. Having done it twice removes the paralysis. Do not skip the copying step even if students are impatient to build their own.

Over-complicating the B section: Students tend to want the B section to be completely different; they start over and lose their groove. Redirect them: remove one element, add one new sound, or change the hi-hat rhythm. Less is almost always more. The best B sections are 80% the same as A.

Volume at maximum: Students default to pushing every fader to the top. Address this on Day 3 with a direct demonstration: play a mix with everything at max, then pull individual elements down and let students hear the difference. The cardinal rule (turn other things down rather than turning one thing up) should be stated explicitly and repeated.

Random hi-hat placement: Students who do not yet understand subdivision will scatter hi-hats without a pattern. Spend 5 minutes at the Day 2 launch explaining the grid: 16 squares = 16 sixteenth notes. Every other square = eighth notes (the default groove feel). Every square = sixteenth notes (busier, more energy).

Opening the piano roll instead of creating a pattern region: This is the most common technical error on Day 2. Watch for students who double-click on a blank space and open an empty piano roll. The correct workflow is right-click (or Control-click), then Create Pattern Region.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

SUGGESTED VARIATIONS ·

More accessible: Remove the AABA structure requirement. Have students focus on building one strong 4-measure A section they are proud of. Introduce AABA in the presentation discussion as a concept to explore next time, not a requirement they failed.

More advanced: Require the B section to be built in 64-step mode (rather than 16). Challenge these students to use note repeat on the hi-hat to create a triplet or 32nd-note accent. Ask them to include in their feedback questions: 'What specific choice makes my B section different, and why does it work?'

GRADING TIPS ·

- Each category is scored 2–5, for a total out of 25. Students self-score first; you score in the second column (page 18).
- Listen once for groove (Composition & Structure), once for contrast between A and B (Creativity), and once with the mixer open (Volume & Balance).
- Technical Skill: open the project. Pattern regions on a clean grid, or 64-step / note repeat used with control, earn the 5.

RESOURCES ·

- The intro deck (.pptx and PDF), included: Day 1 loops lesson feat. Billie Jean, and the Day 2 launch.
- A short "how to identify parts of a drum set" video, and a Logic Step Sequencer tutorial (YouTube; choose ones you like).
- learningmusic.ableton.com/make-beats: a useful conceptual supplement for advanced students.

OBSERVATION NOTES ·

What worked, what to adjust, and anything worth remembering for next year.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING · DAY 1

Digital Music Production 1 · Unit 1

Follow this sequence in order. Do not skip ahead. Steps 3 and 4 we do together as a class.

1. **WATCH** · a short video on the parts of a drum set (your teacher shares the link)
   Learn the names of the main drums before you touch anything.

2. **LABEL** · the Parts of the Drum Set worksheet (page 11) using the word bank at the top
   You should be able to name every piece of the kit without looking by the end.

3. **LOOPS** · the "Beat From Nothing" intro slides, feat. Billie Jean
   What is a loop? Which instruments loop in Billie Jean? Clap the pulse: each clap is a beat, and 4 beats make a measure.

4. **LISTEN** · to the Drumbeat Listening Examples (page 12) and follow along as a class
   Count the kicks. Count the snares. What is the hi-hat doing? What makes each one feel different?

WORDS YOU SHOULD OWN BY THE END OF TODAY

Parts of the kit: kick (bass drum), snare, hi-hat, toms, crash, ride

Loop: a repeated musical idea that builds on what came before

Beat: one clap of the pulse

Measure (bar): 4 beats

Bass: the lowest sound in the mix

If you can define all of these, you're ready for Logic next class.

HELPFUL LINKS ·

- A Logic Step Sequencer tutorial (your teacher shares the link)
- Learning Music beat maker, learningmusic.ableton.com/make-beats: great for understanding beats conceptually

MY QUESTIONS FOR NEXT CLASS

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Parts of the drum set

STUDENT

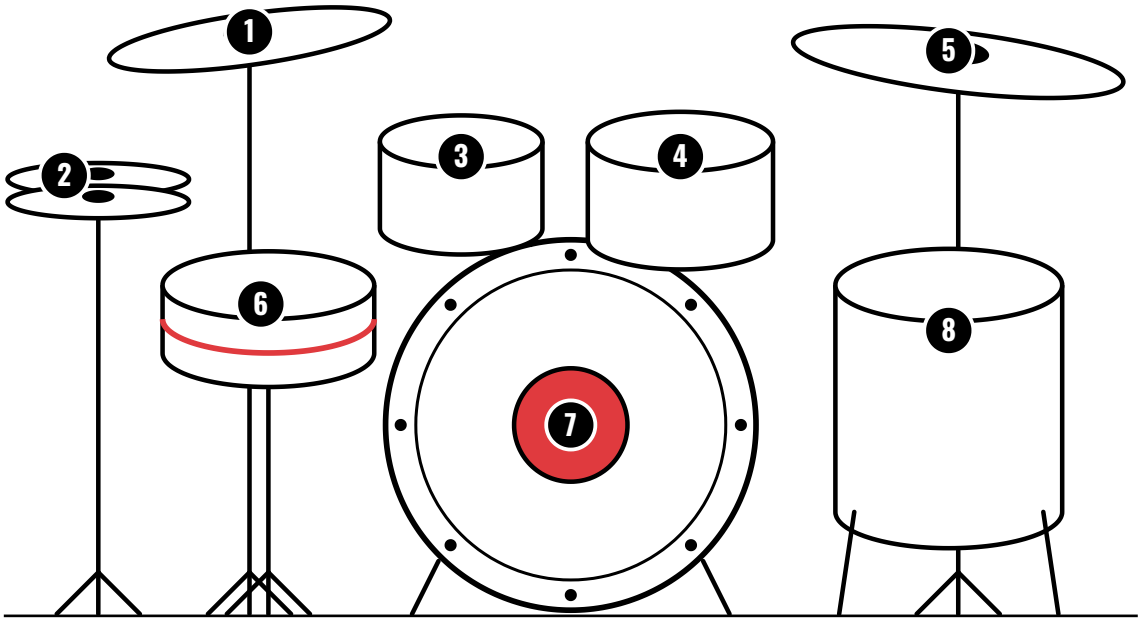
Day 1 · answers p. 19

Name _____ Date _____

Label the parts of the drum set. Write the name from the word bank next to each number.

WORD BANK

HI-HAT | SNARE DRUM | BASS/KICK DRUM | HIGH TOM | MIDDLE TOM | FLOOR/LOW TOM | CRASH CYMBAL | RIDE CYMBAL



1. _____ 2. _____

3. _____ 4. _____

5. _____ 6. _____

7. _____ 8. _____

THINK ABOUT IT

Which three pieces make the basic beat? Which is the lowest sound?

_____

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Drumbeat listening examples

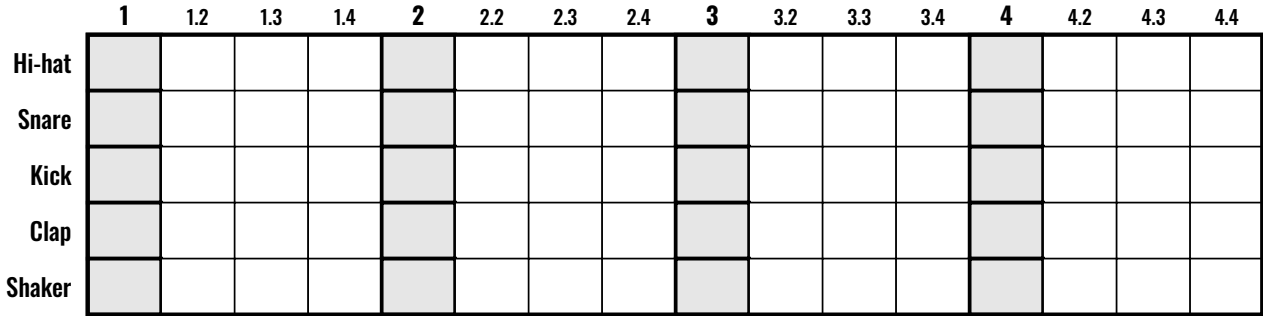
STUDENT

Day 1 · answers p. 19

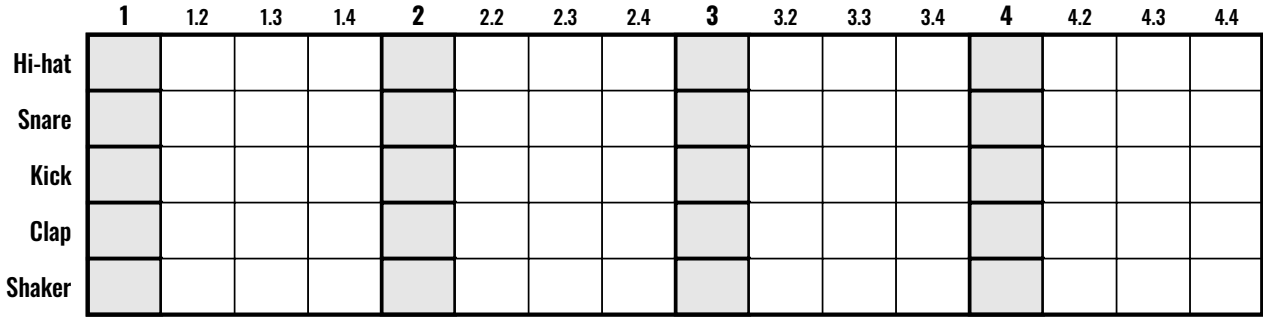
Name _____ Date _____

Your teacher plays three drum beats. Each grid is one measure: 16 squares, four beats (the shaded squares are the beats: 1, 2, 3, 4). Put an X where you hear each sound. Count. Listen. Discuss.

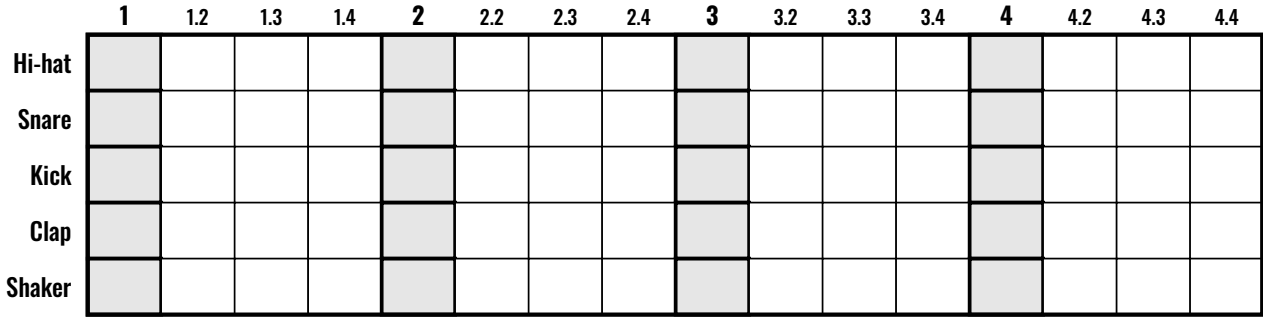
EXAMPLE 1



EXAMPLE 2



EXAMPLE 3



Where does the snare land in all three? _____

Which example feels busiest? Why? _____

What is the hi-hat doing in Example 3? _____

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

STUDENT

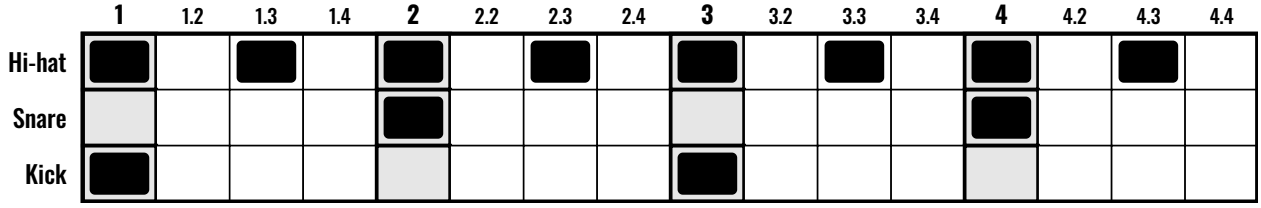
Famous drum patterns

Day 2

Choose ONE pattern and recreate it exactly in Drum Machine Designer. The tempo must match. Each grid is one measure (16 steps). The numbers are Logic's positions: 1 is beat 1, 1.3 is the "and" of beat 1. Call over your teacher when you're done.

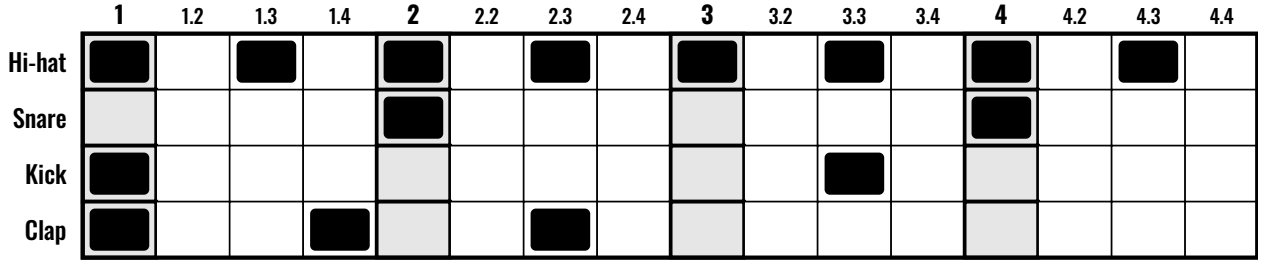
1. BILLIE JEAN · Michael Jackson

Tempo = 116 BPM



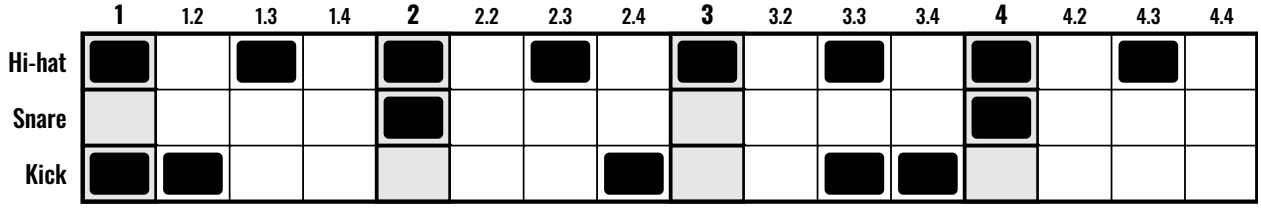
2. CLOSER · The Chainsmokers

Tempo = 96 BPM

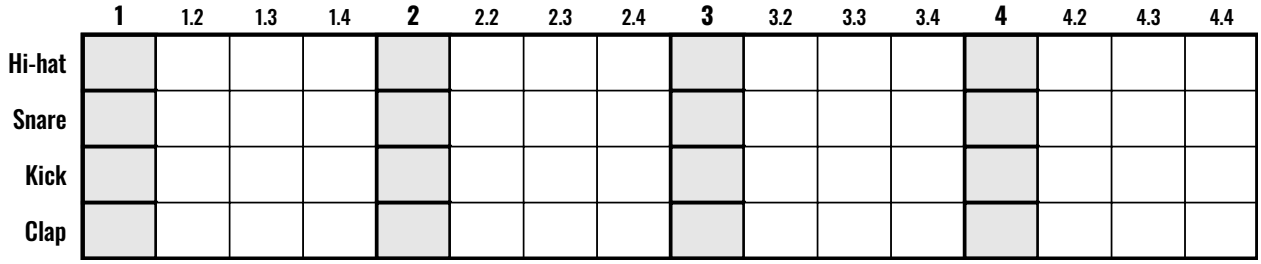


3. WHEN THE LEVEE BREAKS · Led Zeppelin

Tempo = 72 BPM



MY COPY · PATTERN: _____ · TEMPO: ____ BPM



CHECK YOURSELF

- ☐ Every X is on the right square (count from 1)
- ☐ The tempo matches
- ☐ It loops smoothly with Cycle on
- ☐ My teacher checked it: _____

Songs are named for reference only; your teacher can play them from YouTube or Spotify. The grids are simplified teaching charts, not transcriptions.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

PROJECT SHEET

Digital Music Production 1 · Unit 1 · Structured Project · Drum Machine Designer

OVERVIEW ·

You're going to build an original drum beat from nothing: no loops, no pre-made grooves. Starting with the three building blocks of a drum kit (kick, snare, hi-hat), you'll explore rhythm, pattern, and what makes a beat feel right. You'll also analyze a famous drum beat to understand how real producers think about groove. By the end, you'll have made something entirely your own, and you'll understand exactly why it works.

GOALS ·

- Identify the kick, snare, and hi-hat by name, sound, and function
- Build an original drum beat using Logic Pro's Drum Machine Designer
- Understand how volume and panning affect the feel of a mix
- Experience the creative process of building something musical from scratch

REQUIREMENTS ·

- ☐ Built entirely in Drum Machine Designer: no Apple Loops
- ☐ Must include: kick drum, snare drum, closed hi-hat, and at least one additional element (open hi-hat, clap, rim, shaker, etc.)
- ☐ Minimum 16 measures, structured in AABA form (A = main beat, B = variation, A = main beat returns)
- ☐ Volume adjusted thoughtfully on all tracks: nothing at default
- ☐ Panning applied intentionally to at least 2 tracks
- ☐ Project named correctly: Your Name – Beat From Nothing
- ☐ Exported as MP3 and .zip (compressed .logicx file) and submitted to your class page before presenting

YOUR BEAT'S SHAPE



4 measures each · 16 measures · B = A with one change

KEY VOCABULARY

14 terms on page 16: kick, snare, hi-hat, loop, beat, measure, step sequencer, subdivision, tempo, pattern region, cycle, AABA, volume, panning.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Project sheet · steps and tips

STUDENT

STEPS / PROCESS ·

1. **DAY 1 · WATCH & LABEL**
   Watch a short parts-of-the-drum-set video. Label the Parts of the Drum Set worksheet using the word bank. Do not move on until you can name each piece of the kit without looking.

2. **DAY 1 · LOOPS & LISTENING**
   Learn what a loop is using Billie Jean, then follow along with the Drumbeat Listening Examples as a class. Notice where the kick lands, where the snare hits, what the hi-hat is doing. Count. Listen. Discuss.

3. **DAY 2 · COPY A FAMOUS PATTERN**
   Open the Famous Drum Patterns page. Choose ONE pattern and recreate it exactly in Drum Machine Designer; tempo must match. Call over your teacher when you're done.

4. **DAY 3 · BUILD YOUR A SECTION**
   Delete the famous pattern. Now build your own. Start with kick on 1 and 3, snare on 2 and 4, hi-hat in eighth notes. Then make it yours: adjust, add, subtract until it feels right. This is your A section.

5. **DAY 3 · BUILD YOUR B SECTION**
   Create a variation on your A section. You don't need to start over: try removing one element, adding one new sound, or changing the hi-hat rhythm. It should feel related to A but noticeably different.

6. **DAY 3 · STRUCTURE IN AABA**
   Arrange your pattern regions in AABA form: A section, A section, B section, A section. Each section should be 4 measures long for a 16-measure total.

7. **DAY 3 · MIX**
   Adjust the volume of each track so every element has its own space in the mix. Pan at least two elements slightly left or right. Ask yourself: can I clearly hear each drum sound?

8. **DAY 4 · PRESENT**
   Post three feedback questions to the class feedback board before class begins. Share your beat with the class. Respond to at least two of your classmates' projects with genuine, specific feedback.

TIPS FOR SUCCESS ·

- Start simple. A groove that locks in is more powerful than a complicated pattern that doesn't feel right.
- Listen while you build. If you wouldn't nod your head to it, it isn't there yet.
- Your B section doesn't need to be completely different: one small change is often enough to give the listener something new.
- Turn things down before you turn things up. Your mix will almost always sound better when every element has space.
- The hi-hat is what makes a beat breathe. Try eighth notes vs. sixteenth notes and hear how the energy changes.
- When you copy the famous pattern, count carefully. Accuracy here trains your ear to recognize what 'correct' feels like.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

STUDENT

Key vocabulary

14 words

Name _____ Date _____

Read each word and its meaning. In the last column, write an example from your own beat or a song you know.

WORD	MEANING	MY EXAMPLE
Kick drum	the lowest, heaviest drum hit, typically placed on beats 1 and 3	
Snare drum	the sharp, cracking hit that drives the groove, typically on beats 2 and 4	
Hi-hat	a pair of cymbals creating a repeating pulse; can be closed (tight) or open (ringing)	
Loop	a repeated musical idea: the building block of almost every song you hear	
Beat	one steady pulse: what you clap or nod your head to	
Measure (bar)	a group of 4 beats: how music gets counted and organized	
Step sequencer	a grid where you place drum hits across a measure; each square is a subdivision	
Subdivision	how a beat is divided: eighth notes split each beat in 2, sixteenth notes split it in 4	
Tempo	the speed of a song, measured in beats per minute (BPM)	
Pattern region	a looping block of music you build in the step sequencer	
Cycle	Logic's loop playback: repeats a measure or section over and over while you work on it	
AABA form	a structure where A is your main pattern, B is a variation, then A returns	
Volume	how loud a track plays relative to the others in the mix	
Panning	the left-right position of a sound in the stereo field	

QUICK CHECK

A measure has ___ beats. In the Step Sequencer, one measure has ___ squares. _____

Eighth notes: every _____ square. Sixteenth notes: every _____ square. _____

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

Submission checklist

STUDENT

Before you present

Name _____ Date _____

Tick each box before you submit and present. If a box isn't ticked, you aren't done yet.

- [ ] Beat built entirely in Drum Machine Designer: no Apple Loops
- [ ] Kick drum, snare drum, and closed hi-hat are all present
- [ ] At least one additional drum element included
- [ ] AABA structure with a clear, noticeable B section
- [ ] Volume adjusted on all drum tracks: nothing at default
- [ ] Panning applied to at least 2 tracks
- [ ] Project named: Your Name – Beat From Nothing
- [ ] Exported as MP3 (and compressed .logicx) and submitted to your class page
- [ ] Three feedback questions posted to the class feedback board before presenting

MY THREE FEEDBACK QUESTIONS ·

Ask about something you really want to know. Example: "Does my B section feel different enough from my A?"

1. _____
2. _____
3. _____

NOTES ·

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

RUBRIC

Rubric

/ 25

Name _____ Date _____

Score each category from 2 to 5. Score yourself first, then your teacher scores.

Category	2	3	4	5	Self	Teacher
CREATIVITY & ORIGINALITY	Beat is basic with little creative decision-making. Sounds like the default example.	Beat is functional but predictable. B section exists but contrast is minimal.	Beat shows clear creative choices. B section is noticeably different from A.	Beat has a distinct feel and personality. B section creates meaningful contrast. Goes beyond the obvious.		
TECHNICAL SKILL / DAW USE	Basic use of the step sequencer with significant errors or workflow confusion.	Step sequencer used but some elements seem randomly placed. Some workflow confusion.	Step sequencer used correctly. All required elements present and placed with intention.	Drum Machine Designer used confidently. Pattern regions, 64-step, or note repeat used with control.		
COMPOSITION & STRUCTURE	No clear structure. Beat does not repeat in a musical way.	Structure attempted but A and B sections are not clearly differentiated.	AABA structure is present and mostly clear. Beat feels mostly locked in rhythmically.	AABA structure is clear and musically satisfying. Beat locks in and feels complete.		
VOLUME & BALANCE	No meaningful volume or panning adjustments. Everything at default levels.	Some volume adjustments made but mix feels unbalanced. Little or no panning applied.	Most elements are balanced. Volume and panning adjustments are evident and mostly appropriate.	All elements are balanced with intention. Each drum sound has its own space. Panning adds width without distraction.		
DOCUMENTATION & PRESENTATION	Project not submitted or not presented.	Project submitted late or named incorrectly. Presentation made with minimal engagement.	Project named correctly and submitted. Presentation made. Feedback questions posted.	Project correctly named, submitted on time. Confident presentation. Three thoughtful feedback questions posted.		
				TOTAL	**/ 25**	**/ 25**

ONE THING I'M PROUD OF

_____

_____

TEACHER COMMENTS

_____

_____

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

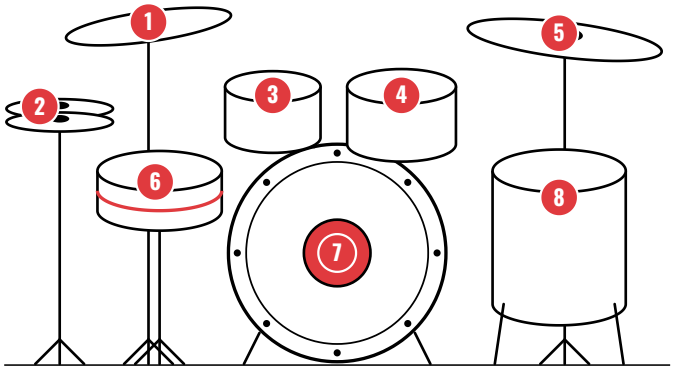
BEAT FROM NOTHING

Answer key · kit labels and listening examples

ANSWER KEY

Teacher

PARTS OF THE DRUM SET



LABELS

1. **Crash Cymbal**
2. **Hi-Hat**
3. **High Tom**
4. **Middle Tom**
5. **Ride Cymbal**
6. **Snare Drum**
7. **Bass/Kick Drum**
8. **Floor/Low Tom**

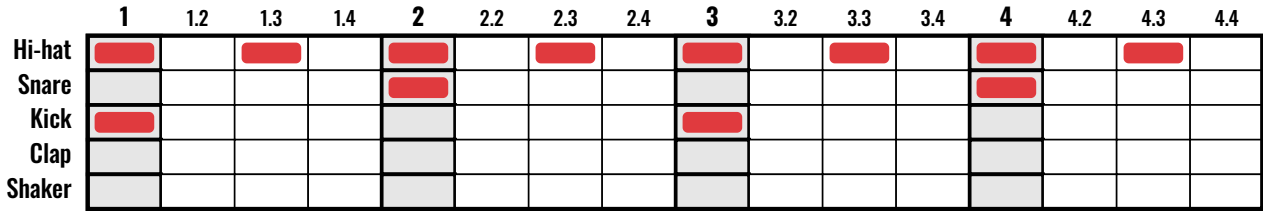
THINK ABOUT IT

Basic beat: kick, snare, hi-hat. Lowest sound: the kick (bass drum).

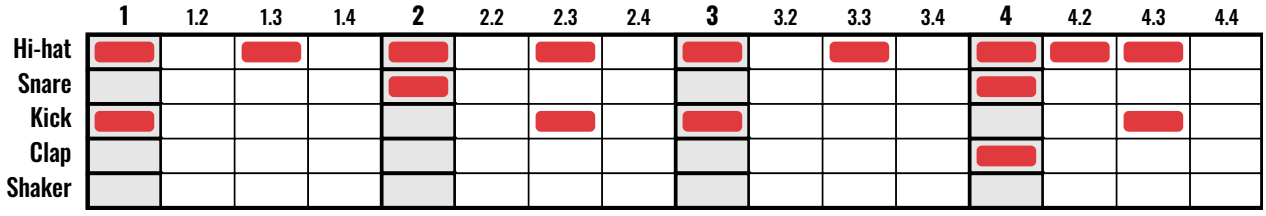
DRUMBEAT LISTENING EXAMPLES ·

Program these in Drum Machine Designer before Day 1, one pattern region each (any tempo around 90–110 BPM works). Example 1 is the same as the Billie Jean pattern.

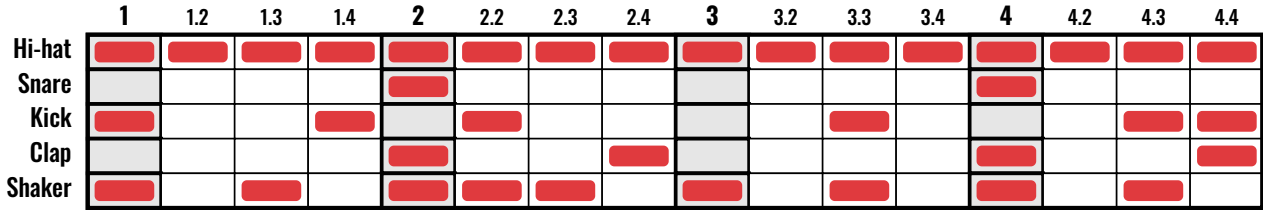
EXAMPLE 1



EXAMPLE 2



EXAMPLE 3



LISTENING QUESTIONS

Snare: beats 2 and 4 in all three. Busiest: Example 3 (sixteenth-note hi-hats, a shaker, a clap that doubles the snare, a syncopated kick). Example 3's hi-hat plays every square: sixteenth notes.

VOCABULARY QUICK CHECK

4 beats; 16 squares. Eighths: every other square. Sixteenths: every square.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

BEAT FROM NOTHING

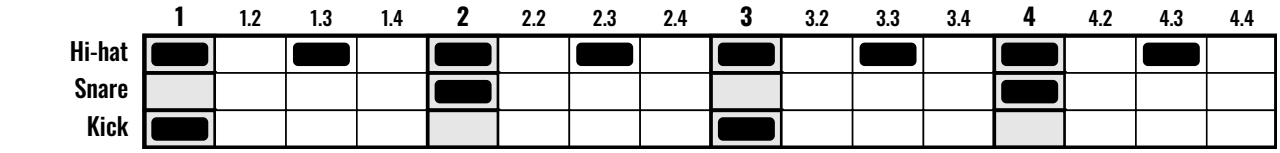
EXAMPLE

Example notes · a worked beat, step by step

Teacher

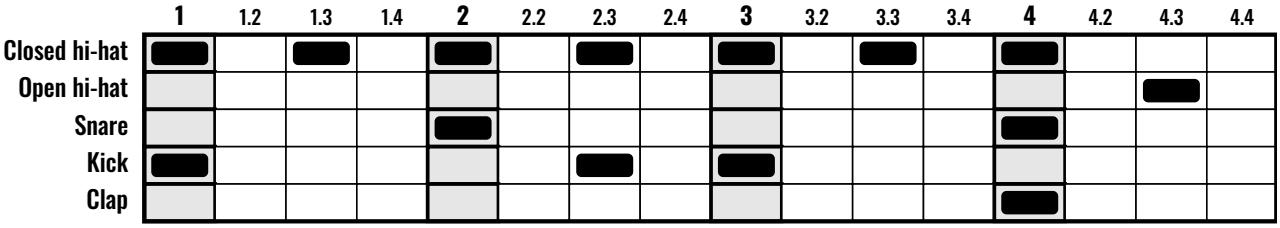
Use this to model the build on Days 2–3 or as a reference while you circulate. It is one possible answer, not the answer: every student's beat should sound different.

1 · STARTING POINT



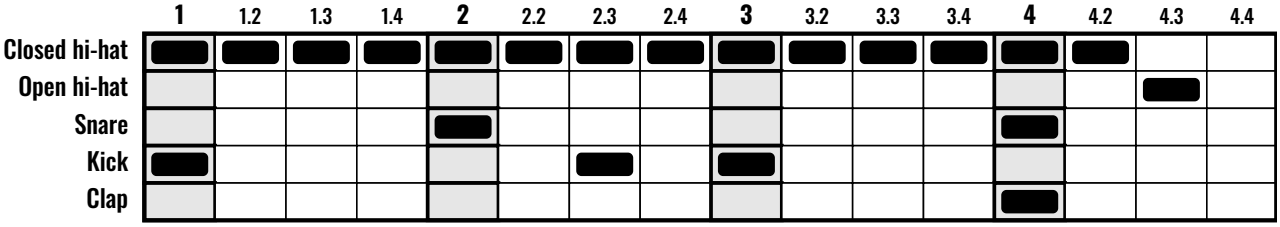
Kick on 1 and 3, snare on 2 and 4, hi-hat in eighths. Every hit lands on the grid, so it locks in. Now change one thing at a time and listen.

2 · MY A SECTION



Extra kick on the "and" of 2 (2.3); clap with the snare on 4; open hat on 4.3, where the closed hat drops out.

3 · MY B SECTION (ONE CHANGE)



Same beat, closed hi-hat in sixteenths: more energy, still clearly related to A.

4 · ARRANGE



4 measures each · 16 measures · B = A with one change

5 · A SAMPLE MIX

TRACK	VOLUME	PAN	WHY
Kick	0 dB (loudest)	Center	The anchor: everything else sits around it.
Snare	about −3 dB	Center	Cuts through on 2 and 4.
Clap	about −6 dB	Slightly left (L10)	Doubles the snare on beat 4; tucked under it.
Closed hi-hat	about −9 dB	Slightly right (R15)	Sits above the kick; never louder than the snare.
Open hi-hat	about −11 dB	Slightly left (L15)	A quick lift into the next bar.

Numbers are starting points for a demo, not targets. Trust your ears: if you can't hear a sound clearly, turn something else down first.

WHAT A 5 SOUNDS LIKE

A groove you nod your head to within a bar, a B section you notice without being told, every drum sound clear, width from panning that doesn't distract, and a student who can say why they made each choice.

© 2026 Brian Lotze · Hudson Beat · For single-classroom use

TERMS OF USE

Thank you for your purchase!

© 2026 Brian Lotze. Hudson Beat by Brian Lotze. All rights reserved. This purchase gives **one teacher** a license to use this resource (the PDF and the intro deck) with their own students.

YOU MAY

- Print and copy pages for your own classroom and students, year after year.
- Present the intro deck to your own classes and edit it for your own teaching.
- Share pages with your own students through a password-protected class site or learning platform.
- Leave the pages for a substitute teacher who teaches your own students.

YOU MAY NOT

- Share, email or copy this resource for other teachers, a whole school or a district. Additional licenses are available on Teachers Pay Teachers.
- Post any part of it on a public website, shared drive or social media.
- Sell, give away or claim any part of it as your own, or edit it into a new product.

QUESTIONS?

Please use the Q&A tab on the Hudson Beat by Brian Lotze store on Teachers Pay Teachers. If something looks wrong, let me know there and I will fix it.

CREDITS

- Curriculum, project sheet, rubric, teacher notes and intro deck: original work by Brian Lotze, Hudson Beat.
- Drum kit and drum machine drawings, grids and layouts: original work, Hudson Beat.
- Text fonts: Oswald (The Oswald Project Authors) and Source Sans 3 (Adobe), SIL Open Font License 1.1.
- Songs are named for reference only (title and artist). No audio, video, lyrics or album art are included. Logic Pro, Drum Machine Designer and Apple Loops are trademarks of Apple Inc., named only to describe compatibility; this resource is not affiliated with or endorsed by Apple. Padlet, YouTube and Spotify are named only as examples of tools you may already use.

Designed with the help of digital and AI tools, and checked by hand.